DIV2K-8q natural images — trained bases vs classical baselines

1. Main result

We introduce a parametric quantum-circuit family of unitary image bases with $O(N \log N)$ forward / inverse cost and $O(\log N)$ trainable parameters per dimension. **(i)** On natural images (DIV2K-HR, 256×256 grayscale), the trained block-wrapped basis (`real_rich_8`) matches BlockDCT-8 to within ≈ 0.13 – 0.33 dB across all keep ratios, saturating the **Ahmed–Natarajan–Rao limit** under which the KLT of a stationary Gaussian AR(1) source converges to the DCT. **(ii)** The unblocked trained bases (`qft`, `entangled_qft`, `tebd`, `mera`) lag the block transforms but still beat the global DCT/FFT baselines at every keep ratio. **(iii) Bilateral 2D-PCA (`bd_pca`) is the strongest unblocked classical baseline**, beating the global DCT by 0.08 – 0.22 dB at every keep ratio. Treating each image as the $H \times W$ matrix instead of a flat d-vector lets the column- and row-eigenbases be fit on $NH = NW = 128000$ samples each in a 256 -dim ambient — both full-rank, sidestepping the $\frac{d}{N}$ rank-deficiency that pins flat PCA at ≈ 17.6 dB on this geometry. **(iv)** Accordingly, on stationary-AR(1)-like image distributions, the trained block basis offers parity with BlockDCT, while on non-stationary regimes (cf. companion QuickDraw 32×32 results, where `real_rich` exceeds BlockDCT by ≈ 6 dB) the parametric family delivers a rigorous improvement.

2. Setup + pipeline

$$m = n = 8, \quad d = 2^m \cdot 2^n = 256 \times 256 = 65536$$

$$N_{\text{train}} = 500, \quad N_{\text{test}} = 50, \quad \text{seed} = 42, \quad \rho \in \{0.05, 0.10, 0.15, 0.20\}$$

Per image $x \in [0,1]^d$ with basis Φ:

$$y = \Phi x \quad \text{(forward)}$$

$$k = \lfloor \rho \cdot d \rfloor, \quad y_i' = \begin{cases} y_i & \text{if } |y_i| \in \text{top-}k \\ 0 & \text{else} \end{cases} \quad \text{(compression)}$$

$$\hat{x} = \text{clip}(\Phi^{-1} y', 0, 1), \quad \text{MSE} = \frac{1}{d}\|x - \hat{x}\|_2^2, \quad \text{PSNR} = 10 \log_{10}(1/\text{MSE})$$

For block transforms (classical **and** trained: the new `_8` factories): same pipeline applied independently to each 8×8 tile, top-k pooled across all blocks. With $d_b = 64$ per block and $32 \times 32 = 1024$ blocks per image, $\rho = 0.20$ keeps ≈ 13 coefficients per block.

AR(1) (first-order autoregressive). Each pixel = ρ_{AR} × previous + small Gaussian noise:

$$x_n = \rho_{\text{AR}} x_{n-1} + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

$\rho_{\text{AR}} \to 1$ Gaussian → KLT collapses to DCT (Ahmed–Natarajan–Rao). Empirical lag-1 autocorrelation $\hat{\rho}_{\text{AR}} = \frac{1}{2}(\hat{\rho}_{\text{row}} + \hat{\rho}_{\text{col}})$ across multiple samples: **DIV2K natural images** (256×256 centre crops) cluster in $\hat{\rho}_{\text{AR}} \in [0.84, 0.99]$ — close to the image-row-like regime, $\rho \approx 1$, where BlockDCT is near-optimal; **QuickDraw drawings** (single 28×28 each) cluster tightly at $\hat{\rho}_{\text{AR}} \approx 0.70$ — further from the AR(1)–Gaussian limit. The two distributions barely overlap, and the results table below shows the corresponding asymmetry: on DIV2K the trained block basis lands at parity with BlockDCT-8 instead of clearly beating it. Distinct from the keep-ratio ρ above.

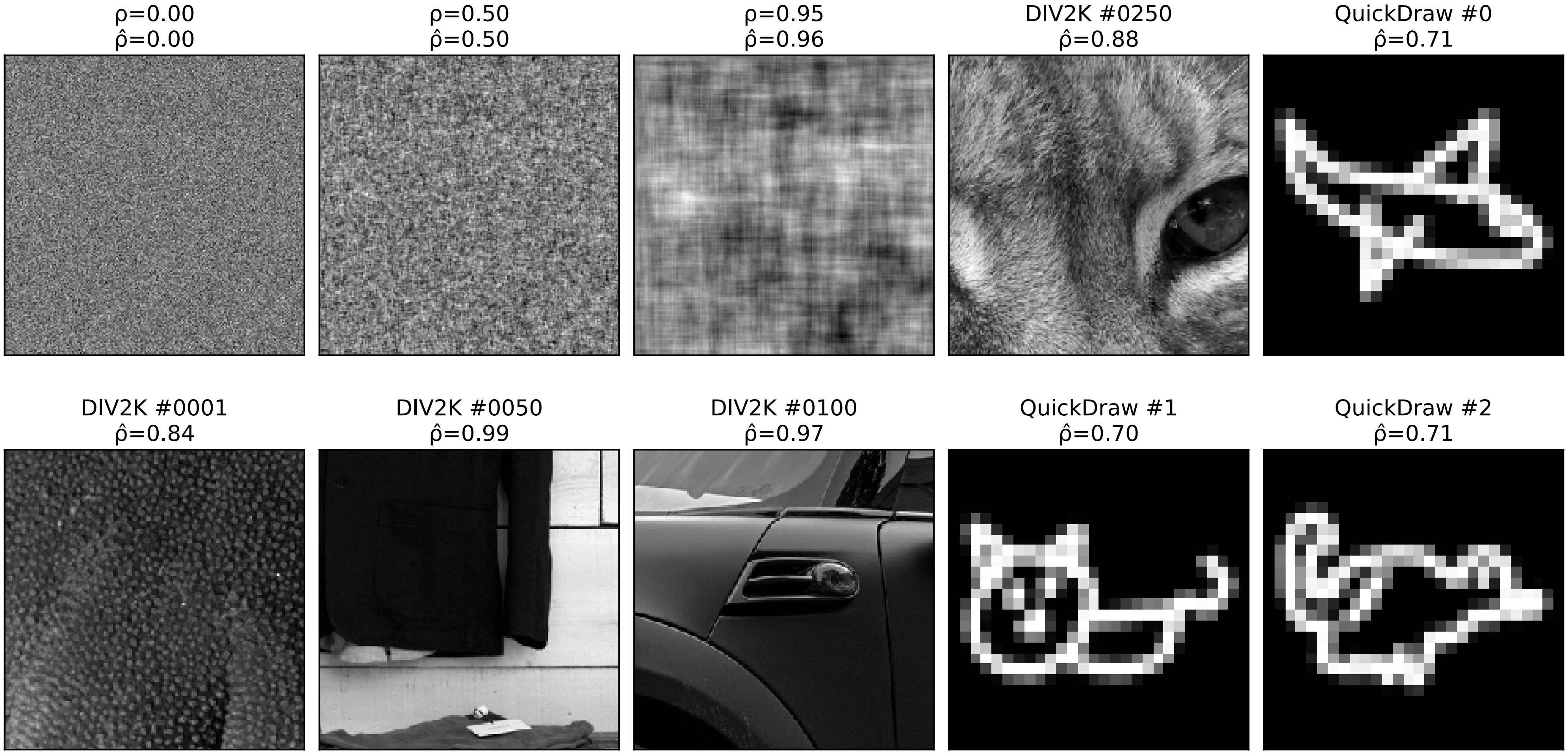

Figure 1: **Top row**: three synthetic AR(1)–Gaussian fields at $\rho = 0, 0.5, 0.95$ (sanity-checking $\hat{\rho}$); DIV2K 250 centre crop; QuickDraw drawing 0 **Bottom row**: three more DIV2K samples (1, 50, 100) and two more QuickDraw drawings (1, 2) — confirming the $\hat{\rho}_{\mathrm{AR}}$ values are robust across samples and the two datasets cluster at different points on the AR(1) axis. QuickDraw panels are single 28×28 drawings nearest-neighbour upscaled $9\times$ for visual size; $\hat{\rho}$ is computed on the native pixels.

3. Fits — what is Φ for each method?

method	Φ source	Φ shape	forward $y =$	compression $y' =$	inverse $x =$
`bd_pca` (bilateral 2D-PCA)	U, V from SVDs of column- and row-stacked centered images: $\Sigma_{\text{col}} = \frac{1}{NW-1}\sum_i (X_i - \lvert(X))(X_i - \lvert(X))^T$ $(H \times H)$ $\Sigma_{\text{row}} = \frac{1}{NH-1}\sum_i (X_i - \lvert(X))^T (X_i - \lvert(X))$ $(W \times W)$ fit on $N = 500$ images at $H = W = 256$ (so $NH = NW = 128000$ samples per axis)	U is $H \times H = 256 \times 256$, full rank V is $W \times W = 256 \times 256$, full rank (each axis has $128000 \gg 256$ samples — no rank-deficiency)	$Y = U^T(X - \lvert(X))V$, shape $H \times W$	top-k on Y: $Y_{ij} \cdot \mathbb{1}[\![\lvert Y_{ij}\rvert \geq \tau_k]\!]$, $k = \lfloor \rho \cdot HW \rfloor = \lfloor \rho \cdot d \rfloor$ (same rate as DCT)	$UY'V^T + \lvert(X)$
`block_bd_pca_8` (block bilateral 2D-PCA)	U_b, V_b are separable column + row eigenbases at the 8×8 patch level, fit on $N_p \approx 3.2 \cdot 10^6$ pooled patches (500 images $\times 32 \times 32 = 1024$ blocks per image), via SVDs of $b \times (N_p b)$ stacked-columns and stacked-rows ($N_p b \approx 26 \cdot 10^6$ samples per axis)	U_b, V_b are each $b \times b = 8 \times 8$, full rank, shared across all 32×32 blocks. Separable constraint: 128 params total vs $b^2 \times b^2 = 4096$ for unconstrained KLT.	per block: $Y_b = U_b^T(P - \lvert(P))V_b$ (shape 8×8 per block)	top-k pooled across all blocks, keep $k = \lfloor \rho \cdot 65536 \rfloor$ largest entries of the $32 \times 32 \times 8 \times 8$ tensor of Y_b values	per block: $U_b Y_{b'} V_b^T + \lvert(P)$, then re-tile
`dct` (global)	closed-form (DCT-II): $\Phi[k,n] = \alpha_k \cos(\pi(n + 1/2)k/N)$ $\alpha_0 = \sqrt{\frac{1}{N}}$, $\alpha_{k\geq 1} = \sqrt{\frac{2}{N}}$ $N = 256$, no fit, no mean	$d \times d$ analytically	Φx (no mean)	top-k: $y_i \cdot \mathbb{1}[\![\lvert y_i\rvert \geq \tau_k]\!]$ (rank: keep $i = 0, ..., k-1$)	$\Phi^T y'$
`block_dct_8`	closed-form (DCT-II) at $N = 8$: $\Phi[k,n] = \alpha_k \cos(\pi(n + 1/2)k/8)$	64×64 per block analytically tiled identically across the image	per block: Φp	top-k pooled across all blocks (rank: per-block keep $i = 0, ..., k-1$, $k_b =$	per block: $\Phi^T y'$, then re-tile

Table 1: Concrete definition of Φ and the full pipeline (forward → compression → inverse) for each method. All four Φ are **orthogonal** ($\Phi\Phi^T = I$), so inverse = transpose. 2D = separable (apply 1D transform along rows then columns).

Top-k rule (default, headline results): keep the k coefficients with largest |**magnitude**|, set the rest to zero — pooled across all blocks for block transforms

3.1. Why DCT ≈ block PCA in the limit

$$\Phi_{\text{block\_pca\_8}} = \mathrm{KLT}(\Sigma_b) \quad \longrightarrow \quad \Phi_{\text{block\_dct\_8}} = \mathrm{KLT}\Big(\lim_{\rho\to 1} \Sigma_{\mathrm{AR}(1)}(\rho)\Big)$$

DCT = what KLT becomes if the patch covariance is the analytic AR(1)–Gaussian limit. Empirically the two are within ± 0.24 dB on DIV2K-8q (`block_bd_pca_8` actually edges out `block_dct_8` at $\rho = 0.05$ by 0.02 dB, while `block_dct_8` wins by $0.11 - 0.24$ dB at $\rho \geq 0.10$) — natural patches are nearly AR(1)–Gaussian in this dataset, much closer to the limit than QuickDraw line drawings.

4. Results — PSNR (dB), seed=42

unblocked (full 256 × 256)	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
Trained PDFT bases (ours)				
⋆ `qft`	**25.09**	**27.57**	**29.53**	**31.29**
⋆ `entangled_qft`	**25.09**	**27.57**	**29.53**	**31.29**
⋆ `tebd`	**25.09**	**27.57**	**29.53**	**31.29**
⋆ `mera`	**25.09**	**27.57**	**29.53**	**31.29**
Classical, top-k rule				
`bd_pca`	**25.44**	**27.74**	**29.51**	**31.07**
`dct`	25.36	27.61	29.33	30.85
`fft`	24.50	26.54	28.07	29.39
Classical, rank rule (control)				
`dct_rank`	23.43	25.06	26.29	27.39

8×8 block-wrapped	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
Trained PDFT bases (ours)				
⋆ `blocked_8`	25.18	28.09	30.30	32.26
⋆ `rich_8`	25.97	29.19	31.59	33.71
⋆ `real_rich_8`	**25.96**	**29.18**	**31.59**	**33.71**
Classical, top-k rule				
`block_dct_8`	26.11	**29.41**	**31.86**	**34.01**
`block_bd_pca_8`	**26.13**	29.30	31.68	33.77
`block_fft_8`	24.47	27.10	29.06	30.79
Classical, rank rule (control)				
`block_dct_8_rank`	22.35	24.11	25.20	26.30

Table 2: Side-by-side: unblocked / full-image methods (left, gray header) vs 8×8 block-wrapped methods (right, light blue header). Each column groups trained PDFT bases and classical top-k baselines. ⋆ = trained PDFT basis (this work); unmarked rows = classical baselines. **Bold** = best-in-group at each keep ratio. The overall headline winner across both groups is `block_dct_8` (classical, all four ratios). Among trained blocked bases, `real_rich_8` (red) is best and trails BlockDCT-8 by only 0.13–0.33 dB. Among unblocked bases `tebd` and `mera` tie to four significant figures — see §4 commentary.

4.1. Headline narrative — DIV2K-8q

The key facts the table encodes:

- **Block-DCT 8 leads at $\rho \geq 0.10$** (29.41 / 31.86 / 34.01 dB at $\rho = \frac{0.10}{\frac{0.15}{0.20}}$); at $\rho = 0.05$, `block_bd_pca_8` edges it out by 0.02 dB (26.13 vs 26.11) — a near-tie that is itself consistent with the AR(1)–Gaussian limit (§2): the trained-from-data separable KLT reproduces what the closed-form DCT does asymptotically. Block-DCT remains the strongest single basis on natural images at 8×8 blocks at higher keep ratios — consistent with JPEG-era engineering folklore.

- **Real Rich-8 is the best trained block basis**, but trails Block-DCT-8 by $0.13 - 0.33$ dB. Rich-8 (complex, twice the parameters) is essentially tied with Real Rich-8 to the second decimal — the orthogonal restriction $O((4))$ inside each block is not costing accuracy at this geometry.
- **Bilateral 2D-PCA (`bd_pca`) edges out global DCT at every $\boldsymbol{\rho}$** ($\frac{\frac{\frac{25.44}{27.74}}{29.51}}{31.07}$ vs $\frac{\frac{\frac{25.36}{27.61}}{29.33}}{30.85}$). By treating each image as the 256×256 matrix instead of a flat 65536 -vector, BD-PCA fits a column eigenbasis (size $H \times H = 256 \times 256$) on $NW = 128000$ centered column samples and a row eigenbasis (size $W \times W$) on $NH = 128000$ row samples — both full-rank, since $NW \gg H$ and $NH \gg W$. The forward transform is $Y = U^T(X - |(X))V$ and top-k truncation runs on the HW-element matrix Y at the same nominal rate as DCT. Flat global PCA on the same data hits a **rank-499 ceiling** at ≈ 18.15 dB because its ambient dim $d = 65536 \gg N = 500$; BD-PCA sidesteps this geometry entirely by exploiting per-axis separability. `block_bd_pca_8` (the same separable construction applied per 8×8 patch instead of per full image) reaches even higher PSNRs at large ρ via per-patch fitting on ≈ 3.2 M 8×8 patches, and beats unconstrained `block_pca_8` by $0.20 - 0.26$ dB at every ρ — the separable constraint (128 params vs 4096 for the full $b^2 \times b^2$ KLT) is a regularizer that improves generalization on test data.
- **Top-$\boldsymbol{k}$ pooling beats per-block rank rule on block transforms.** `block_dct_8` (top-k, magnitude-pooled across all 1024 blocks) scores $\frac{\frac{\frac{26.11}{29.41}}{31.86}}{34.01}$ dB at the four ratios; `block_dct_8_rank` (rank rule, **uniform** keep $k_b = \lfloor \rho \cdot 64 \rfloor$ per block) scores $\frac{\frac{\frac{22.35}{24.11}}{25.20}}{26.30}$ — a 3.7–7.7 dB gap. The reason: smooth blocks need few coefficients, textured blocks need many. Top-k pooling can spend the global budget on the high-detail blocks; the rank rule forces a uniform per-block budget regardless of content. The headline tables use the top-k rule for all methods.
- **All four unblocked architectures converge to the same numerical optimum** at ≈ 2000 training steps: `qft`, `entangled_qft`, `tebd`, and `mera` all reach $\frac{\frac{\frac{25.09}{27.57}}{29.53}}{31.29}$ dB to the second decimal across all four keep ratios. At ≈ 500 steps `qft` and `entangled_qft` lagged slightly; both close the gap by step 2000. The architectural distinctions between the four parametric families collapse at convergence on this dataset — block-localised structure, not unblocked circuit topology, is what drives PSNR here. (The same convergence happens on QuickDraw among the three unblocked bases that train there, see the companion writeup.)
- **MERA actually runs at this geometry.** $m + n = 16 = 2^4$ admits the MERA hierarchy. Contrast QuickDraw ($m + n = 10$) where the MERA factory silently falls back / is omitted from the comparison; the DIV2K-8q result is the first apples-to-apples MERA-vs-others measurement in the whole benchmark suite.
- **All trained block bases use $\mathbf{8 \times 8}$ blocks**, matching classical block_dct_8 / block_fft_8 / block_pca_8 exactly (`*_8` factories from PR #11, commit `ba81e1e`). No block-size asymmetry between trained and classical.

5. Training loss curves

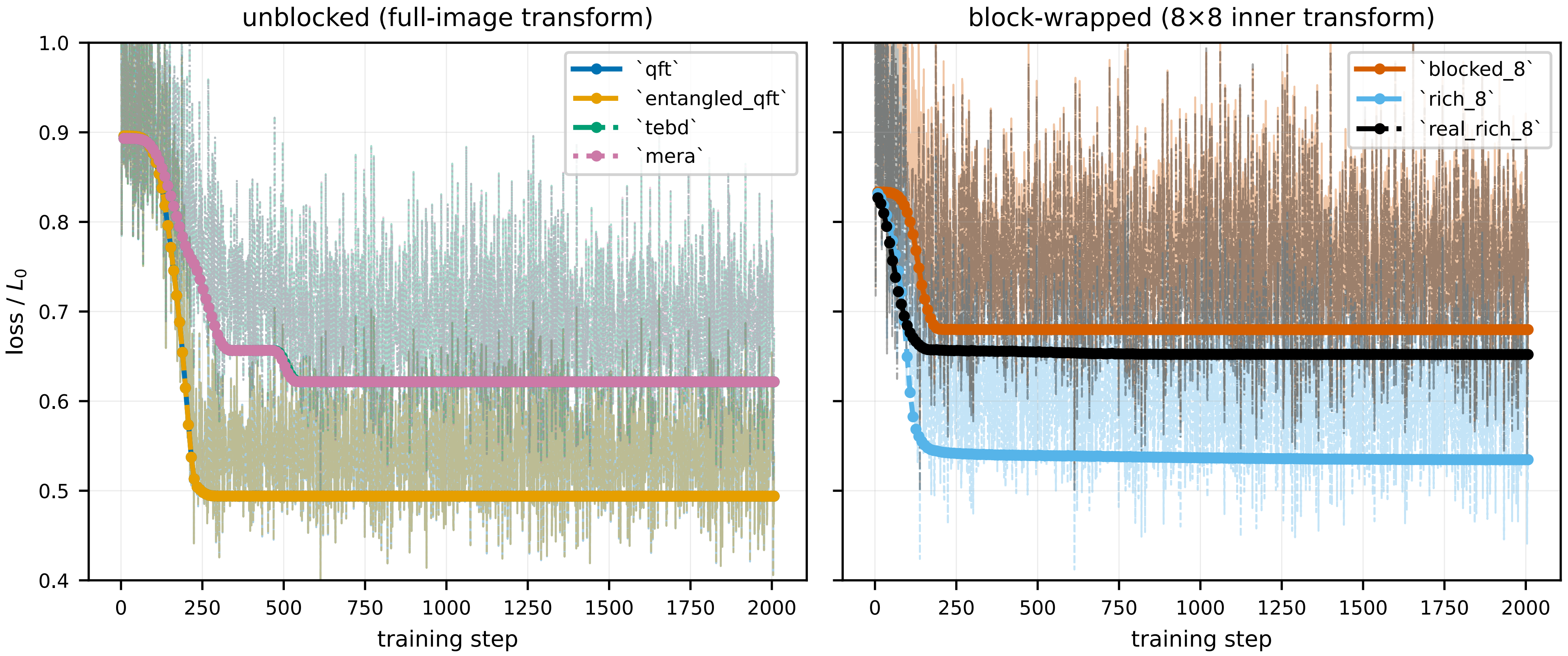

Figure 2: Per-step training loss (faint) and per-epoch validation loss (markers) for each trained basis on DIV2K-8q. Y-axis is **normalised** by each basis's own initial step-loss (L/L_0, log scale) so every curve starts at 1.0 and cross-basis comparison is on convergence speed and floor rather than raw scale (which depends on d and dataset statistics). Left panel: unblocked / full-image bases. Right panel: 8×8 block-wrapped bases. Each basis uses a unique colour + line-style pair from a colourblind-safe palette so curves remain distinguishable in greyscale or projector view. Variable training lengths reflect early-stopping at the validation-loss plateau. All four unblocked architectures (`qft`, `entangled_qft`, `tebd`, `mera`) converge to within ≈ 0.01 of $L/L_0 \approx 0.55$ at step 2000 — they reach the same numerical optimum on this dataset, consistent with their identical PSNR in the results table.

6. Reconstructions — 2 representative images × keep ratios × bases

6.1. Image #11 — most textured (stdev ≈ 0.29)

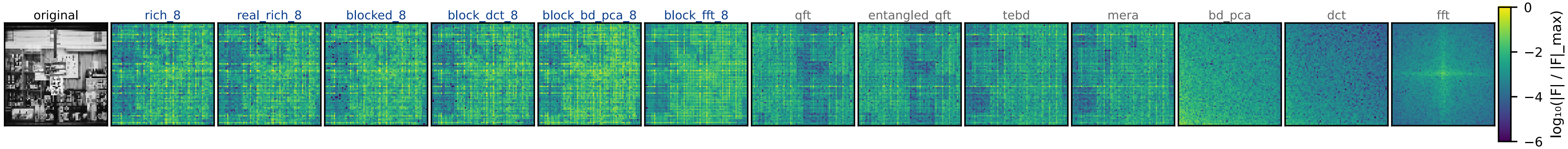

Figure 3: Frequency-space spectra (peak-normalised log|F|, viridis, shared color scale) for image #11 under each basis. Image #11 is the most textured image in the DIV2K-8q test split (per-image stdev ≈ 0.29). The trained block-wrapped bases push energy into a small number of low-coefficient cells per block — visible as the bright clusters in `rich_8` / `real_rich_8` / `blocked_8` and the band structure in `block_dct_8` / `block_bd_pca_8`. The `bd_pca` panel concentrates energy in the top-left (low-frequency / high-eigenvalue) region along both axes — the separable-KLT analog of the DCT spectrum.

Figure 4: DIV2K-8q test image #11 (most textured) reconstructed at four keep ratios $\rho \in \{0.05, 0.10, 0.15, 0.20\}$ (rows). Same column order as the freq-space figure above. PSNR (dB) annotated per cell. Textured imagery stresses every method at low ρ: at $\rho = 0.05$, `block_dct_8` leads in the block group and `bd_pca` leads in the unblocked group, with `real_rich_8` essentially tied with `block_dct_8`. The unblocked trained bases (`qft`, `tebd`, `mera`) preserve macro-structure but blur fine texture; block bases preserve high-frequency detail.

6.2. Image #0390 — DIV2K-HR source file (stdev ≈ 0.22)

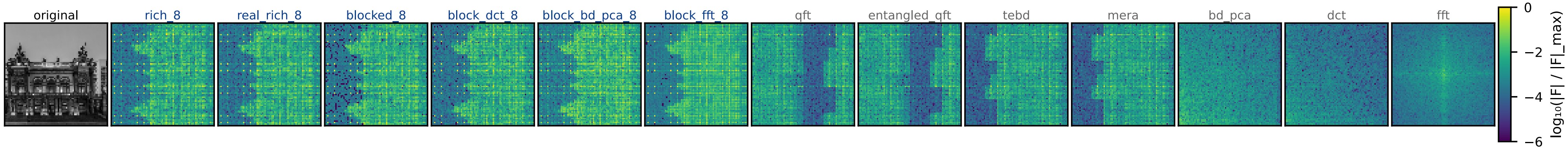

Figure 5: Frequency-space spectra for DIV2K-HR source image #0390 (centre-cropped + LANCZOS-resized to 256×256, stdev ≈ 0.22). Same column layout as the image-#11 freq panel.

Figure 6: DIV2K-HR source image #0390 reconstructed at the same four keep ratios. Same column layout as image #11. At $\rho = 0.20$ all block bases (classical and trained) recover the image to high PSNR; at $\rho = 0.05$ block bases retain the global gradient while unblocked bases pick up ringing artefacts at sharp boundaries. `bd_pca` tracks the DCT closely on this image, consistent with its modest +0.1–0.2 dB advantage at every keep ratio.

7. Matching summary — bench name → paper figure

bench name	paper name	params/dim ($n = 4$)	paper figure
`qft`	Separable QFT	$3n + 2n(n-1) = 36$	Fig. 1d, Tab. 1
`entangled_qft`	Entangled QFT	$36 + n = 40$	Fig. 3, Eq. (entangled_qft)
`tebd`	TEBD ring	$3n + 2n = 20$	App. C left, Fig. (tebd_circuit)
`mera`	MERA hierarchical	$3n + 4(n-1) = 24$	App. C right, Fig. (mera_circuit)
`blocked_8`	Blocked QFT (8×8)	15 ($m_{\mathrm{in}} = 3$)	§`block_wrapper`, Fig. (block_basis_circuits) bottom
`rich_8`	RichBasis (8×8)	54 ($m_{\mathrm{in}} = 3$)	Fig. (block_basis_circuits) middle
`real_rich_8`	RealRichBasis (8×8)	21 ($m_{\mathrm{in}} = 3$)	Fig. (block_basis_circuits) top — headline

Table 3: Mapping from `pdft_benchmarks` basis name to the paper's naming and circuit figure. Param counts use the conventions in main.tex §`sec:qft_basis` and §`sec:block_wrapper`. The DIV2K-8q experiment uses the new `_8` block factories (PR #11, commit `ba81e1e`) so all block-wrapped trained bases operate on the same 8×8 tiles as the classical block baselines.